

Sam Kay AMIMA

MATHEMATICIAN · MATHEMATICS OUTREACH & COMMUNICATION · FREELANCE MUSICIAN

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Personal profile

As a highly motivated and dedicated student with a deep passion for learning and personal growth, I consistently go above and beyond in my studies. I am recognised for my exceptional work ethic and efficient time management skills, which have enabled me to consistently meet deadlines and attain outstanding academic results. I possess an unwavering aspiration to succeed and am always endeavouring to elevate myself to new heights of accomplishment and knowledge.

I am agreeable, enthusiastic and a clear communicator. A skill I am most proud of is the ability to convey complex ideas with a flexible vocabulary, tailoring conversations to a wide variety of audiences.

Research experience

Doctor of Philosophy (PhD) in Applied Mathematics

Durham, UK

Durham University

Oct 2025 – Present

- Working towards a PhD in solar physics under the supervision of Dr Peter Wyper.
- Developing models of various structures in the solar wind, falling under the realm of analytics in magnetohydrodynamics (MHD).

Master of Mathematics (MMath)

Durham, UK

Durham University

Sept 2021 – July 2025

- Graduated with Master's degree in mathematics with First Class Honours.
- Involved with the Mathematical Society.
- **Selected modules:** Advanced Quantum Theory, Fluid Mechanics, General Relativity, Geophysical & Astrophysical Fluids, Superstrings.

Outreach programs

City of Maths Leeds

Leeds, UK

Event day ambassador

Jan 2026 – Present

- Ambassador for the UK's first *City of Maths* project, aiming to enthral the nation with maths and how it can be brought into everyday life.
- Events include the *Big Number Natter*, a stall at *Lark in the Park* and various ASDA shopping challenges.
- [Click here to learn more and get involved!](#)

Chalkdust magazine

Remote

Article writer and editor

Feb 2024 – Present

- Writing and editing articles for a popular maths magazine, distributed internationally and among universities.
- Occasional cryptic crossword setter under the setter name *Seuss*.
- [Click here to read the most recent issue.](#)

24 Hours of Le Maths

DUMPSS

Lead event organiser

Dec 2025 – Mar 2026

- Organised the university's fifth annual 24-hour maths endurance event to raise money for *Point North*.
- Managed a team of volunteer question writers and markers; liaised directly with the charity.
- [Click here for the 2026 question paper.](#)

Chalkboard Ultra podcast

DU MathSoc

Podcast co-host and lead editor

Oct 2023 – June 2025

- Hosting a podcast that brings light to all areas of mathematics, with conversations among PhD students and lecturers.
- [Click here for the Spotify playlist.](#)

Skills

Programming $\LaTeX$, TikZ, Python, HTML, Markdown, Maple, Desmos, VisualPDE, ARMS, Heliospace.

Soft Skills Teamwork, problem solving, organisation, data analytics, critical thinking, report writing, presentation skills.

Miscellaneous Microsoft Excel, Canva, sleight of hand and card magic.

Research projects

Modelling magnetic switchbacks

Oct 2025 – Present

PhD research project

- Developing analytic models for magnetic switchbacks: rapid reversals in the Sun's ambient magnetic field that would otherwise be radial.
- Will continue this research through MHD simulations via ARMS and compare results to see if we can describe these phenomena.

The Standard Model from a Symmetry Perspective

Oct 2024 – May 2025

MMath research project

- Researched the construction of the Standard Model of Particle Physics; studied gauge theory and spontaneous symmetry breaking.
- Deepened my understanding through reading academic papers and lecture notes from various sources.
- Produced a report of my research, alongside an academic poster and presentation, [available to view on my website](#).

On Modelling Seismic Waves

Dec 2022 – Jan 2023

Mathematical Modelling module

- Conducted research into how we can simulate the effects of seismic waves to better understand the Earth's structure.
- A solo project, programming 2-dimensional partial differential equation solvers into Python.

Work experience

Undergraduate tutorial leader

Durham, UK

Durham University

Oct 2025 – Present

- Assisting undergraduate students with their studies in the form of tutorials; guiding them through chosen problem sheet questions and being a mentor.
- Teaching assistant for Fluid Mechanics III; tutorial leader for Mathematical Methods II and Methods of Mathematical Physics II.

Summer camp counsellor & Group leader

Starrucca, Pennsylvania

Island Lake Camp

Summers 2023 – 2026

- Mentor, counsellor and head of for children aged 6–16 at a sleep-away summer camp in America.
- Working as a part of the STEM, Outdoor Adventure and Rockshop departments, ensuring children were safe and engaged in their activities.

Maths and science tutor

Remote

MyTutor.co.uk // Freelance

Jan 2022 – June 2025 // Present

- Previously tutoring in a classroom environment via video call and private tutoring with 100% return business.
- Now private tutoring on a word-of-mouth basis.

Volunteering

Maths subject representative

Stephenson College

Stephenson College JCR

Jan 2025 – Present

- Point of guidance contact for students regarding mathematical sciences.
- An advisory role for those needing help with exam techniques, module choices and general academic problems.

Jazz band president

Josephine Butler College

Butler Jazz & Butler Soul Band

Sept 2023 – Nov 2024

- Leading two bands after being the lead pianist for two years and Librarian for one year prior.
- Directing rehearsals, booking gigs, organising charity events, overseeing the executive team, managing band finances.
- Improved skills in performance, public speaking, organisation and networking with other bands.

Interests

Outreach Associate member of the Institute of Mathematics and its Applications (IMA).

Fitness Regularly attend the gym and enjoy running; enjoy long walks in unspoilt countryside.

Performing Music Regular at Durham Jazz Society's jazz jam nights @ Fabio's bar and Stag's Open Mic @ the Stag's Head.

Music Learning the guitar and piano since age eight, now playing mostly jazz and funk music. Avid fan of the keytar.

References available upon request

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